

There are a total of six problems. You have to solve all of them.

Problem 1 (CO5): Pumping Lemma (5 points)

Let $\Sigma = \{0, 1\}$. Consider the following language.

$$L = \{w \in \Sigma^* : w = 0^a 1^b 1^c 0^d, \text{ where } a + b = c + d \text{ and } a, b, c, d \geq 0\}$$

Use the pumping lemma to **demonstrate** that L is not regular language.

Problem 2 (CO3): Designing Context-Free Grammars (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages. Recall that for a string w , $|w|$ denotes the length of w .

$$L_1 = \{w \in \Sigma^* : w \text{ is an even length palindrome}\}$$

$$L_2 = \{w \in \Sigma^* : \text{length of } w \text{ is even}\}$$

$$L_3 = \{x1y : x, y \in L_2, |x| = |y|\}$$

$$L_4 = L_1 \cap L_3$$

Now solve the following problems.

- Give a context-free grammar for the language L_1 . (3 points)
- Give a context-free grammar for the language L_3 . (4 points)
- Give a context-free grammar for the language L_4 . (3 point)

Problem 3 (CO4): The CYK Algorithm (5 points)

Apply the CYK algorithm to fill up the table for the string $aaaba$ using the following grammar. Here a and b are terminals, and the rest are variables.

$$\begin{aligned} S &\rightarrow BA \mid BC \\ A &\rightarrow AB \mid AC \mid a \\ B &\rightarrow CB \mid CC \mid b \\ C &\rightarrow CA \mid a \end{aligned}$$

1,5				
{?}				
1,4	2,5			
{S, A, B, C}	{?}			
1,3	2,4	3,5		
{?}	{A, B, C}	{?}		
1,2	2,3	3,4	4,5	
{A, B, C}	{A, B, C}	{?}	{S}	
1,1	2,2	3,3	4,4	5,5
{A, C}	{A, C}	{A, C}	{B}	{A, C}
a	a	a	b	a

Problem 4 (CO3): Constructing Pushdown Automata (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages.

$$L_1 = \{w \mid w \text{ starts and ends with the same character}\}$$

$$L_2 = \{w \mid \text{the number of 0s in } w \text{ is not same as the number of 1s}\}$$

- Give the state diagram of a pushdown automaton that recognizes L_1 . (4 points)
- Give the state diagram of a pushdown automaton that recognizes L_2 . (6 points)

Problem 5 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Take a look at the grammar below and solve the following problems.

$$\begin{aligned} A &\rightarrow 1A \mid 1C \mid 0B \mid 00A \\ B &\rightarrow 0A \mid 1B \mid 00B \\ C &\rightarrow 0C0 \mid 0C1 \mid 1C0 \mid 1C1 \mid \epsilon \end{aligned}$$

- Give a leftmost derivation for the string 01011001 . (3 points)
- Sketch the parse tree corresponding to the derivation you gave in (a). (2 points)
- Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string. (3 points)
- Find a string w of length six such that w has exactly one parse tree in the grammar above. (1 point)
- Design an unambiguous Context Free Grammar for the language represented by the given ambiguous grammar. (1 point)

Problem 6 (CO4): Chomsky Normal Form (10 points)

Answer the following questions.

- List the productions that violate the conditions of the Chomsky Normal Form (CNF) in the following grammar. (5 points)

$$\begin{aligned} M &\rightarrow \epsilon \mid x \mid MN \\ N &\rightarrow NC \mid yy \mid P \\ P &\rightarrow QQ \mid z \\ Q &\rightarrow yQ \mid \epsilon \end{aligned}$$

- Write down the additional rules that need to be added to the following grammar if the production $B \rightarrow \epsilon$ is removed. (3 points)

$$\begin{aligned} S &\rightarrow Aa \mid BB \\ A &\rightarrow CA \mid BaB \mid \epsilon \\ B &\rightarrow AB \mid b \mid \epsilon \end{aligned}$$

- Write down the additional rules that need to be added to the following grammar if all the unit productions are removed. (2 points)

$$\begin{aligned} X &\rightarrow 0X1Y1 \mid Y \\ Y &\rightarrow XY \mid YY \mid Z \\ Z &\rightarrow 1Z \mid Z \end{aligned}$$